

1. (a) Solve the system:
$$\begin{cases} x + y = 12 \\ x - y = 4 \end{cases}$$

(b) Solve for x and y : $2x + 3y = 18$,
 $4x - y = 8$
2. The inverse of an invertible matrix always exists.
(A) True
(B) False
3. The determinant of a diagonal matrix equals the product of its diagonal entries. from Q5
(A) True
(B) False
4. Answer True or False for each of the following statements:
 - (a) The determinant of a triangular matrix is equal to the product of its diagonal entries.
(A) True
(B) False
 - (b) If a matrix A is invertible, then its determinant must be zero.
(A) True
(B) False
 - (c) The trace of a matrix is always equal to the sum of its eigenvalues.
(A) True
(B) False
 - (d) The transpose of a symmetric matrix is the negative of itself.
(A) True
(B) False